2.Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

To write program for electric field and electric flux density in sphere using MATLAB.

Test case (i)

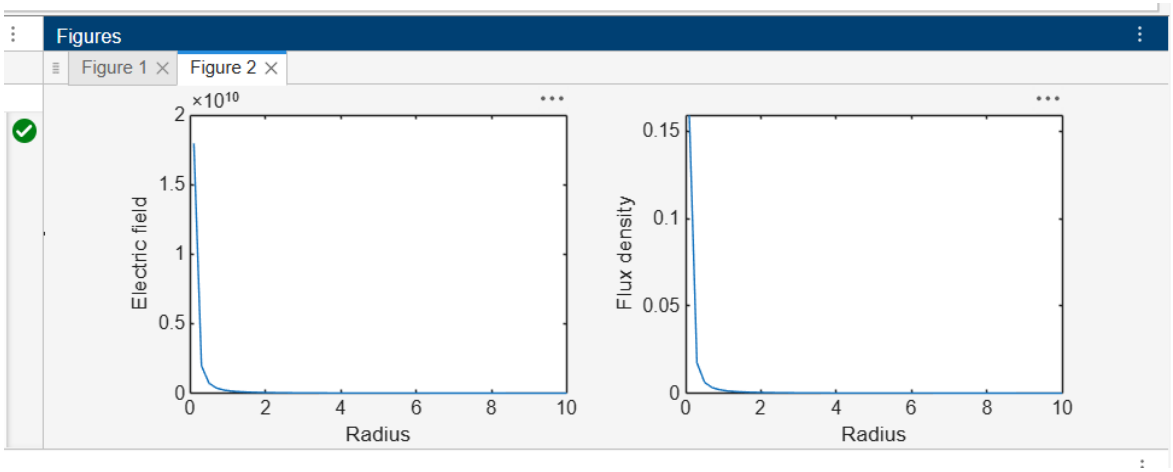
Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius

Program:

```
clc;
clear;
q=input('Enter the value of q=');
r=input('Enter the value of r=');
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
E=k*q/(r^2);
disp("The electric field is");
disp(E);
D=epsilon*E;
disp("The flux density is");
disp(D);
% Plot
x=linspace(0.1,10,50);
E_plot=k*q./(x.^2);
D_plot=epsilon*E_plot;
figure;
subplot(1,2,1);
plot(x,E_plot);
xlabel('Radius');
ylabel('Electric field');
```

```
subplot(1,2,2);
plot(x,D_plot);
xlabel('Radius');
ylabel('Flux density')
```

OUTPUT:



Test case (ii)

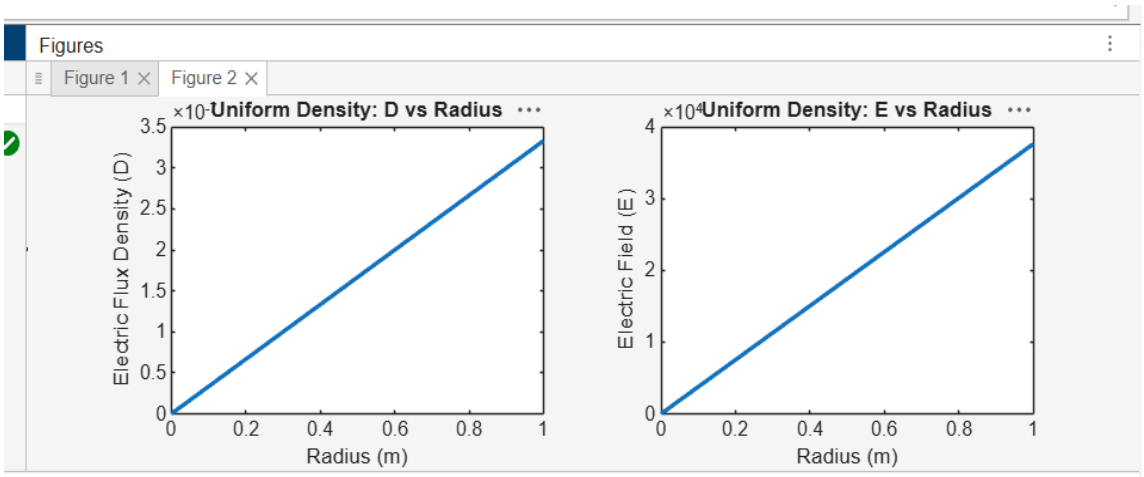
Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ0

Program:

```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
r = linspace(0, R, 100);
D = (rho0 .* r) / 3;
E = D / epsilon;
subplot(1,2,1);
plot(r,D,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Flux Density (D)");
title("Uniform Density: D vs Radius");
subplot(1,2,2);
```

```
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Uniform Density: E vs Radius");
```

OUTPUT:



Test case (iii) For a sphere with a linearly increasing charge density ρv = ρ0r, derive and plot the electric field (E) and electric flux density (D)

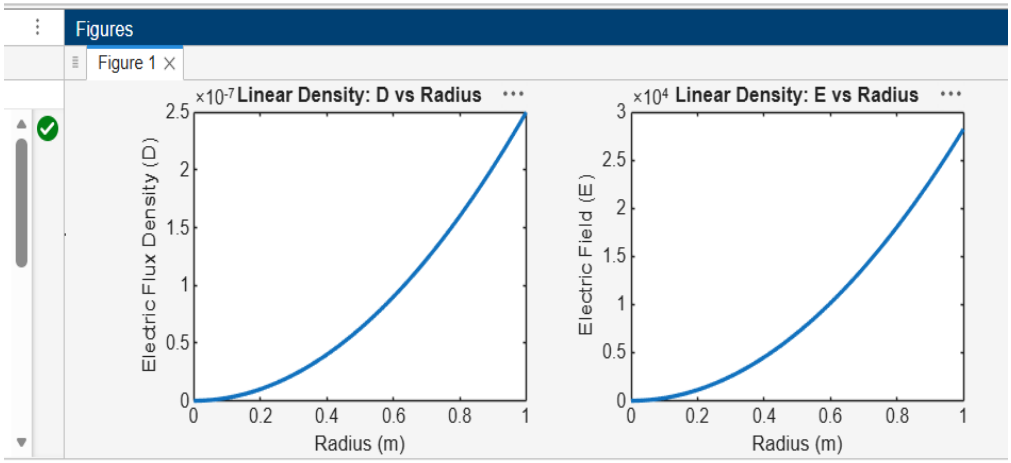
Program:

```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
r = linspace(0, R, 100);
D = (rho0 .* r.^2) / 4;
E = D / epsilon;
subplot(1,2,1);
plot(r,D,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Flux Density (D)");
title("Linear Density: D vs Radius");
subplot(1,2,2);
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
```

ylabel("Electric Field (E)");

title("Linear Density: E vs Radius");

OUTPUT:



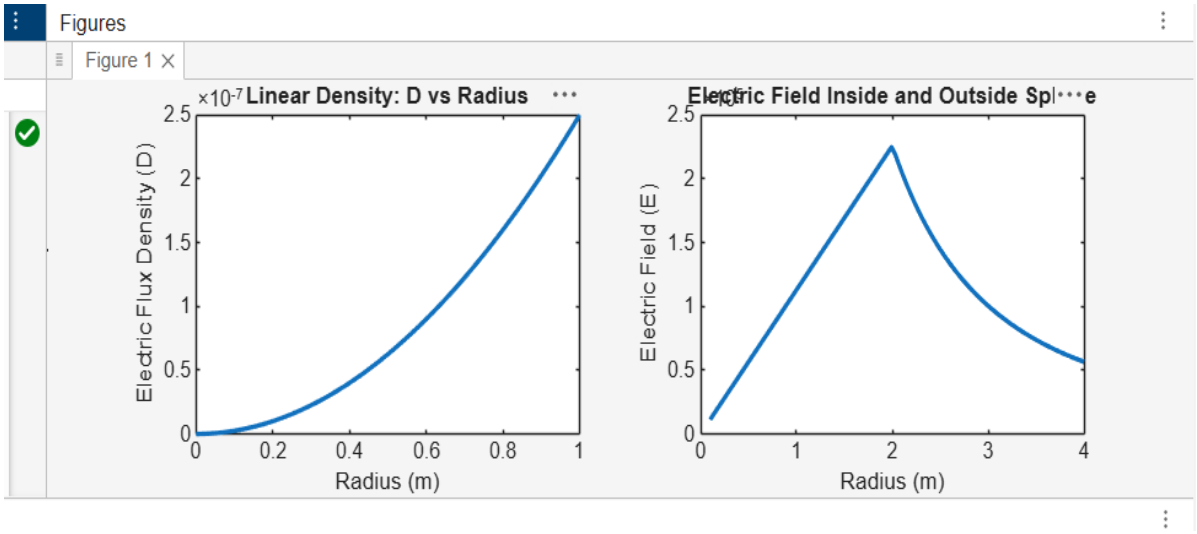
Test case (iv) A sphere of radius R = 2 m has a uniform charge density ρ = 3×10−6 C/m3.Calculate and plot electric field inside and outside the sphere

Program:

```
clc;
rho0 = 3e-6;
epsilon = 8.854e-12;
R = 2;
r = linspace(0.1, 4, 100);
E = zeros(size(r));
for i = 1:length(r)
    if r(i) < R
        E(i) = rho0 * r(i) / (3 * epsilon);
    else
        Q = rho0 * (4/3) * pi * R^3;
        k = 1/(4*pi*epsilon);
        E(i) = k * Q / (r(i)^2);
    end
end
plot(r,E,'LineWidth',2);
```

```
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Electric Field Inside and Outside Sphere");
```

OUTPUT:



Result:

Thus the program was verified for electric field and electric flux density in sphere using SCILAB was successfully

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Select the appropriate charge-distribution model	Select the appropriate charge-distribution model	Correctly distinguishes point, uniform-volume and specified non-uniform charge models and their assumptions.	Selects the correct model with minor clarification needed.	Shows partial understanding; multiple corrections are needed.	Uses an unsuitable model or assumptions.	25	2.5	
2	Implement transformations in MATLAB	Implement six transformations in MATLAB	Implements all six transformations correctly using suitable	Implements most transformations correctly with minor	Implements some transformations but needs assistance to	Unable to implement the required transformations.	25	2.5	

			degree-based functions.	coding errors.	correct errors.				
3	Verify numerical values and reverse transformations	Implement E and D calculations in MATLAB	Correctly calculates electric field E and flux density D with consistent SI units.	Calculations are mostly correct with minor coding or unit errors.	Some calculations work but require assistance.	Unable to obtain valid E and D values.	30	3	
4	Analyze radial variation and validate results	Analyze radial variation and validate results	Plots the correct inside/outside or stated distribution behaviour and verifies against theory.	Plots and validation are mostly correct with minor discrepancies.	Plots or theoretical validation are incomplete.	Plots are incorrect or results are not validated.	20	2	
Total							100	10	